

Visual, Auditory and Tactile Preventive Maintenance 230925

1. Early Detection and Dynamic Anomalies (Running Controls)

Why running? Many failures only manifest under real load, temperature or operational speed.

Visual Controls: Allow to identify:

Abnormal vibrations or movements (which cannot be identified when stopped).

Active wear or leaks of aluminum, sand, cement, pigment, water, lubricant or fluid.

Irregular movement patterns (polycords, belts, rubber bands, discs, chains, shafts).

Auditory Controls: The human ear is excellent for detecting subtle changes:

Pitch changes: Rollers, bands, bearing shafts, pulley belts, reducers/gears, etc.

Knocking or metallic sounds indicating clearances or impacts.

Justification: A stopped equipment is "sleeping" equipment.

These operational inspections are the first and fastest alarm system.

2. Status Verification and Static Integrity Verification (Stopped Controls)

To access, measure and verify what is impossible or dangerous with equipment running.

Manual/Tactile Controls:

Pulley dirt: Assess pulley dirt level.

Clearance adjustment: Axial/radial play in shafts, level of torsional displacement.

Tension verification of chains, belts or polycords.

Wear detection by touch (e.g.: surface roughness, coupling play).

Cleaning and close inspection of critical components.

Justification: These controls provide specific numerical data (e.g.: "2mm clearances, elongations") essential for equipment condition monitoring and planning future interventions.